Khagatara Health Institute (KHI)

Technical Integration Specification: ABDM Sandbox, M1/M2/M3 FHIR R4 & NHCX PM-JAY Pipeline

SYSTEM TARGET ARCHITECTURE OVERVIEW

This specification details the self-hosted, zero-licensing digital health integration engine for KHI deployed on Oracle Cloud Always Free Tier (Ampere A1 ARM Architecture) leveraging the **khagatara.com** domain infrastructure for ABDM Milestone compliance (M1, M2, M3) and National Health Claims Exchange (NHCX) PM-JAY automated claim submissions.

1. DOMAIN & ENDPOINT ROUTING MATRIX

Subdomain / URL	Target Hosting & Environment	Functional Responsibility	Protocol / SSL
khi.khagatara.com	Frontend Web Portal / Vercel Edge	Patient Registration, ABHA Printing, Dialysis Booking	HTTPS / TLS v1.3
abdm.khagatara.com	Oracle Cloud Compute (Gateway)	NHA Sandbox Asynchronous Callbacks & Webhook Bridge	HTTPS / Let's Encrypt
api.khagatara.com	Oracle Cloud Compute (FastAPI)	FHIR R4 Generation, HPR/HIP API, NHCX Claims Engine	HTTPS / Port 443

2. ENTERPRISE INTEGRATION ARCHITECTURE



3. PRODUCTION INFRASTRUCTURE SETUP (ORACLE CLOUD)

Execute the following commands on the Oracle Always Free Tier Ubuntu 22.04 LTS instance to configure NGINX reverse proxy with automated Let's Encrypt SSL certificates:

```
# 1. Update system dependencies & install NGINX + Certbot
sudo apt update && sudo apt install nginx certbot python3-certbot-nginx -y

# 2. Create NGINX Configuration for ABDM Gateway Endpoint
sudo nano /etc/nginx/sites-available/abdm.khagatara.com

# Configuration Payload:
server {
    server_name abdm.khagatara.com api.khagatara.com;

    location / {
        proxy_pass http://127.0.0.1:8000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}
```

4. CORE FASTAPI ABDM GATEWAY ENGINE IMPLEMENTATION

The following production Python service runs on Oracle Cloud handling session authentication token fetching, ABHA verification (M1), and HIP consent webhooks (M2/M3):

```
import os, requests, time
from fastapi import FastAPI, Header, HTTPException, Request, BackgroundTasks
from pydantic import BaseModel

app = FastAPI(title="KHI ABDM & NHCX Integration Engine", version="1.0.0")

ABDM_CLIENT_ID = os.getenv("ABDM_CLIENT_ID", "YOUR_SANDBOX_CLIENT_ID")
ABDM_CLIENT_SECRET = os.getenv("ABDM_CLIENT_SECRET", "YOUR_SANDBOX_CLIENT_SECRET")
AUTH_URL = "https://dev.abdm.gov.in/gateway/v0.5/sessions"

token_cache = {"token": None, "expires_at": 0}

def get_abdm_access_token():
    if token_cache["token"] and time.time() < token_cache["expires_at"]:
        return token_cache["token"]

    payload = {"clientId": ABDM_CLIENT_ID, "clientSecret": ABDM_CLIENT_SECRET}
    response = requests.post(AUTH_URL, json=payload, headers={"Content-Type": "application/json"})

    if response.status_code == 200:
        data = response.json()
        token_cache["token"] = data["accessToken"]
        token_cache["expires_at"] = time.time() + data.get("expiresIn", 1400)
        return token_cache["token"]
    raise HTTPException(status_code=500, detail="ABDM Session Authentication Failed")

@app.get("/health")
def health_check():
    return {"status": "ACTIVE", "facility": "Khagatara Health Institute", "hip_id": "KHI_TVM_01"}

@app.post("/v0.5/users/auth/on-init")
async def abha_auth_on_init(request: Request):
    """Callback endpoint for NHA Sandbox Async Responses"""
    body = await request.json()
    print(f"[ABDM Callback Received] RequestID: {body.get('requestId')} | Status: SUCCESS")
    return {"status": "ACKNOWLEDGED"}
```

5. ABDM MILESTONE 1 (M1): ABHA CREATION & VERIFICATION SCHEMA

Endpoint schema for initiating Aadhaar/Mobile OTP-based ABHA ID generation directly from KHI registration desk:

```
class GenerateAadhaarOtpRequest(BaseModel):
    aadhaar: str

@app.post("/api/v1/abha/generate-aadhaar-otp")
def generate_aadhaar_otp(data: GenerateAadhaarOtpRequest):
    token = get_abdm_access_token()
    headers = {
        "Authorization": f"Bearer {token}",
        "X-CM-ID": "sbx",
        "Content-Type": "application/json"
    }
    # ABDM Sandbox V3 Aadhaar OTP Endpoint
    url = "https://healthidsbx.abdm.gov.in/api/v1/registration/aadhaar/generateOtp"
    res = requests.post(url, json={"aadhaar": data.aadhaar}, headers=headers)
    return res.json()
```

> **ABDM SANDBOX INTEGRATION COMPLIANCE NOTE**

6. ABDM MILESTONE 2 & 3: FHIR R4 BUNDLING ENGINE

KHI generates standardized **FHIR R4 DiagnosticReport and Bundle JSON** records for every dialysis and laboratory session to adhere to National Health Data Specifications:

```
def build_fhir_dialysis_bundle(patient_abha_id: str, patient_name: str, session_id: str) -> dict:
    return {
        "resourceType": "Bundle",
        "id": f"KHI-BUNDLE-{session_id}",
        "meta": {"lastUpdated": "2026-10-20T10:00:00.000+05:30"},
        "type": "document",
        "entry": [
            {
                "fullUrl": f"Patient/{patient_abha_id}",
                "resource": {
                    "resourceType": "Patient",
                    "id": patient_abha_id,
                    "name": [{"text": patient_name}],
                    "gender": "male"
                }
            },
            {
                "fullUrl": f"DiagnosticReport/DIALYSIS-{session_id}",
                "resource": {
                    "resourceType": "DiagnosticReport",
                    "id": f"DIALYSIS-{session_id}",
                    "status": "final",
                    "code": {
                        "coding": [{
                            "system": "http://loinc.org",
                            "code": "10982-3",
                            "display": "Hemodialysis care plan"
                        }]
                    },
                    "subject": {"reference": f"Patient/{patient_abha_id}"},
                    "conclusion": "Dialysis session completed without complications. Ultrafiltratio
n: 2.5L."
                }
            }
        ]
    }
```

7. PM-JAY / NHCX NATIONAL CLAIMS EXCHANGE PIPELINE

Automated payload submission structure for processing hemodialysis claims under PM-JAY package codes over the NHCX network:

```
@app.post("/api/v1/nhcx/submit-claim")
def submit_pmjay_claim(patient_abha: str, pmjay_card_id: str, package_code: str = "KMC001"):
    """
    KMC001: PM-JAY Package Code for Hemodialysis Session
    """
    token = get_abdm_access_token()
    claim_payload = {
        "resourceType": "Claim",
        "identifier": [{"system": "https://nhcx.gov.in/claim", "value": f"KHI-CLAIM-{int(time.time
())}"}],
        "status": "active",
        "type": {"coding": [{"system": "http://terminology.hl7.org/CodeSystem/claim-type", "code":
"institutional"}]},
        "use": "claim",
        "patient": {"identifier": {"system": "https://healthid.abdm.gov.in", "value": patient_abh
a}},
        "provider": {"identifier": {"system": "https://hpr.abdm.gov.in", "value": "KHI_TVM_01"}},
```

8. VERIFICATION & NHA SANDBOX EXIT PROTOCOL

Step #	Verification Action Item	Testing Tool / Portal	Acceptance Criterion
Step 1	Sandbox Registration & Credentials	`sandbox.abdm.gov.in`	Client ID & Secret issued for KHI
Step 2	Webhook Callback Reachability	Certbot / Postman	`https://abdm.khagatara.com` SSL 200 OK
Step 3	Milestone 1 (M1) Functional Test	ABDM Sandbox Suite	Successful ABHA Creation & OTP Verify
Step 4	Milestone 2 & 3 (M2/M3) FHIR Push	FHIR Validator Tool	Valid FHIR R4 Bundle schema match
Step 5	NHCX Mock Claim Submission	NHCX Sandbox Portal	ACK generated with valid Claim ID
Step 6	Production Key Application	NHA Developer Support	Production API Key issued to KHI

9. OPERATIONAL MAINTENANCE & MONITORING CHECKLIST

- **Oracle Cloud Systemd Service:** Run FastAPI backend as a persistent background daemon via `systemctl enable khi-backend`.
- **Automated Certbot Renewal:** Ensure SSL certificate renewal cronjob runs bi-weekly (`certbot renew --quiet`).
- **Token Refresh Caching:** Keep ABDM session authentication token cached locally in memory to prevent exceeding NHA API rate limits.
- **Audit Logging Compliance:** Store encrypted access logs locally on Oracle 200 GB block storage for 3 years as mandated by DISHA/ABDM data privacy norms.

> **FINAL EMPANELMENT AUTHORIZATION SIGN-OFF**
>
> This technical integration document serves as the official architecture reference for Khagatara Health Institute during NHA ABDM Sandbox Exit Approval and PM-JAY / NHCX Hospital Empanelment with the Kerala State Health Agency (SHA).